

REAL-WORLD SETUP EXAMPLES		
Room Type	Projector Model	Key Setup Details
Compact Apartment Living Room	Yaber K300s UST Laser	Placed 9 cm from wall on low console; paired with 100" ALR screen; autofocus and auto-keystone for quick setup.
Dedicated Home Theater Room	Hisense C2 Ultra 4K Laser DLP	Ceiling-mounted at 4.5 m for 150" screen; used ISF Dark Room mode; multi-point focus and professional calibration.
Multi-Purpose Living Space	Epson Home Cinema 5050UB 4K LCD	Shelf-mounted with manual keystone correction; 2,500 ANSI lumens to handle ambient light; used 1.3 gain grey screen.

They perform best when projector and ambient light sources are not co-linear and can boost perceived contrast by 30–50% in bright rooms.

Indian Supplier Example: ELCOR Pro-Series DIY ALR screens (120", tab-tensioned) are available for ₹2.9 lakh, optimized for UST laser projectors.

3) Ultra-short-throw (UST) screen considerations

UST projectors require specialized ALR screens with downward-facing microstructures (e.g., Liberty Screen Pro 100" floor-rising tab-tensioned ALR) to handle extreme light angles without hotspots.

Placement, mounting, and installation

1) Ultra-short-throw in compact spaces

Laser UST models like the Yaber K300s cast 100" images from under 25 cm away, featuring autofocus and auto-keystone for hassle-free setup – ideal for apartments or multi-use rooms.



2) Ceiling vs. shelf mount

- **Ceiling Mount:** Provides permanent, out-of-the-way installation; use adjustable mounts and ensure proper airflow clearance.
- **Shelf/Console Placement:** Requires manual keystone correction or lens shift; suitable for portable use but may disrupt room aesthetics.

Step-by-step ceiling mount:

1. **Locate Joists:** Use a stud finder to secure mount into structural supports.
2. **Measure Drop:** Calculate mount extension so lens aligns with screen center.
3. **Install Mount Plate:** Drill pilot holes and anchor securely.
4. **Attach Projector:** Level and tighten screws; verify lens shift and zoom range.
5. **Cable Management:** Route HDMI, power, and control cables through conduit or raceways.

Image alignment and keystone correction

Once your projector is mounted and powered on, precise image alignment ensures a geometrically accurate, distortion-free picture.

1. Lens shift (If available):

Use vertical and horizontal lens-shift controls to move the image without tilting the projector, preserving optical clarity.

2. Digital keystone correction:

- Engage digital keystone only when lens shift is insufficient. Adjust vertical and horizontal keystone to square the image to the screen edges.
- **Caution:** Excessive digital correction can degrade resolution; limit keystone use to <10° where possible.

3. Corner correction (Advanced models):

High-end projectors (e.g., Hisense C2 Ultra) offer individual corner adjustment via on-screen menus, allowing fine-grained correction of trapezoidal distortion.

4. Check for screen edge fit:

Project a grid or geometry test pattern (often supplied via HDMI sources or built-in menus) to ensure the image aligns perfectly with the screen border.

5. Focus and uniformity:

- Focus the center of the screen first, then adjust four focus “zones” (one in each quadrant) if your projector supports multi-zone focus.